Computer Networks — Lecture 6 (Ch 2, Lec 3): Cookies, Web Caching, HTTP/2 & HTTP/3

Questions Bank — بنك الأسئلة

ملخص البنك: 64 سؤال اختيار من متعدد (MCQ) + 16 سؤال مقالي (Essay) + 5 سؤال حسابي (Calculation). توزيع الإجابة الصحيحة في الـ MCQ: A=16 / B=15 / C=16 / D=17. الأسئلة مرتبة بحيث الـ high-likely ييجوا الأول في كل نوع.

Part 1 — Multiple Choice Questions (MCQ)

1.

HTTP uses cookies because the HTTP server is:

ترجمة: HTTP بيستخدم cookies لأن السيرفر بيتوصف بـ إيه؟

A. Reliable

B. Persistent

C. Connection-oriented

D. Stateless

الإجابة الصحيحة: D) Stateless

ليه صح: HTTP server stateless — ما بيبقاش عنده معلومات عن الـ clients. cookies بتحل المشكلة دي عشان تـ track المستخدمين.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies Stateless **ترجيح: high**

.2

Cookie technology has how many components?

ترجمة: تقنية cookies فيها كام مكوّن؟

A. Two

B. Three

C. Four

D. Five

الإجابة الصحيحة: C) Four

ليه صح: 4 مكونات: cookie header في الـ response، cookie header في الـ request، cookie file على جهاز اليوزر، back-end database على السيرفر. نقطة مرقمة مهمة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (Figure 2.10)

Cookies Components **ترجيح: high**

.3

Which of the following is NOT a component of cookie technology?

ترجمة: إيه اللي مش مكون من مكونات الـ cookies؟

A. A back-end database at the Web site

B. A cookie header line in the HTTP request message

C. A cookie header line in the TCP handshake

D. A cookie header line in the HTTP response message

الإجابة الصحيحة: C) A cookie header line in the TCP handshake

ليه صح: الـ 4 مكونات هي: header في الـ response، header في الـ request، file على اليوزر، database على السيرفر. مفيش cookie header في الـ TCP handshake.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (Figure 2.10)

Cookies Components **ترجيح: high**

.4

Which HTTP response header is used by the server to send a cookie to the browser?

ترجمة: إيه الـ HTTP response header اللي السيرفر بيبعته للـ browser عشان يبعت cookie؟

A. If-Modified-Since:

B. Last-Modified:

C. Set-cookie:

D. Cookie:

الإجابة الصحيحة: C) Set-cookie:

ليه صح: Set-cookie: في الـ response من السيرفر. Cookie: في الـ request من الـ browser. Last-Modified و If-Modified-Since للـ conditional GET.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies | Set-cookie | Headers | ترجيح: high

.5

Which HTTP request header does the browser use to send a stored cookie back to the server?

ترجمة: إيه الـ HTTP request header اللي الـ browser بيستخدمه عشان يبعت cookie للسيرفر؟

A. Set-cookie:

B. Host:

C. User-agent:

D. Cookie:

الإجابة الصحيحة: D) Cookie:

ليه صح: Cookie: في الـ request. مثال: Cookie: 1678. Set-cookie: بيتحط في الـ response، مش الـ request.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies | Cookie header | ترجيح: high

.6

Where is the cookie file kept?

ترجمة: فين الـ cookie file بيتخزّن؟

A. On the user's end system, managed by the user's browser

B. On the origin server only

C. On the Web cache

D. Inside the HTTP response message

الإجابة الصحيحة: A) On the user's end system, managed by the user's browser

ليه صح: على جهاز اليوزر (end system) وبيديره الـ browser. مش على السيرفر ولا على الـ cache ولا جوه رسالة HTTP.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies | Cookie file | **ترجيح: high**

.7

When Susan visits Amazon for the first time, what does the Amazon Web server do?

ترجمة: لما Susan تزور Amazon أول مرة، السيرفر بيعمل إيه؟

A. Stores the cookie directly inside the HTTP request

B. Creates a unique identification number and an entry in its back-end database indexed by that number

C. Rejects the request because Susan has no cookie yet

D. Asks Susan to register with full name and address

الإجابة الصحيحة: B) Creates a unique identification number and an entry in its back-end database indexed by that number

ليه صح: السيرفر بيعمل unique ID وentry في الـ back-end database، ويبعتها في Set-cookie header. مش بطلب تسجيل فوراً ولا برفض.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (Susan/Amazon example)

Cookies | Back-end database | **ترجيح: high**

.8

What does the browser do when it receives a Set-cookie: header in an HTTP response?

ترجمة: إيه اللي الـ browser بيعمله لما يستقبل Set-cookie header في الـ response؟

A. Sends the cookie to all servers it knows
B. Discards the cookie after the session
C. Stores the cookie in the origin server's database
D. Appends a line to the special cookie file it manages, including the hostname and the identification number

الإجابة الصحيحة: D) Appends a line to the special cookie file it manages, including the hostname and the identification number

ليه صح: بيـ append سطر للـ cookie file فيه hostname + ID. مش بيـ discard ولا بيـ broadcast على كل السيرفرز.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies | Browser behavior | **ترجيح: high**

.9

What is the primary reason cookies are considered controversial?

ترجمة: ليه cookies بيتعتبر عليها جدل؟

A. They are slow and increase latency
B. They require upgrading the access link
C. They can be considered an invasion of privacy
D. They are unreliable and cause data loss

الإجابة الصحيحة: C) They can be considered an invasion of privacy

ليه صح: لأنها ممكن تعتبر انتهاك privacy — السيرفر يقدر يتعلم كتير عن اليوزر ويبيع معلوماته لطرف تالت. مش لأنها بطيئة ولا غير موثوقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (privacy paragraph)

Cookies | Privacy | **ترجيح: high**

.10

Amazon uses cookies to provide which service?

ترجمة: Amazon بيستخدم cookies عشان يوفّر إيه؟

A. Shopping cart service

B. File transfer

C. DNS resolution

D. E-mail routing

الإجابة الصحيحة: A) Shopping cart service

ليه صح: shopping cart: بيحتفظ بقائمة مشتريات Susan عشان تدفع collectively في الآخر. DNS و الإيميل و FTP مالهم علاقة بـ Amazon cookies.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies Shopping cart **ترجيح: high**

.11

A Web site can learn a lot about a user by combining:

ترجمة: الموقع يقدر يتعلم كتير عن اليوزر عن طريق دمج إيه مع إيه؟

A. Cookies and TCP sequence numbers

B. Cookies and ARP tables

C. Cookies and user-supplied account information

D. Cookies and DNS records

الإجابة الصحيحة: C) Cookies and user-supplied account information

ليه صح: cookies + معلومات الحساب اللي اليوزر بيدخلها (الاسم، الإيميل، البطاقة). DNS و ARP و TCP sequence مالهم علاقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (privacy paragraph)

Cookies Privacy **ترجيح: high**

.12

A Web cache is also called a:

ترجمة: الـ Web cache بيتسمى كمان؟

A. Origin server

B. Proxy server

C. Mail server

D. DNS server

الإجابة الصحيحة: B) Proxy server

ليه صح: proxy server. بيـ satisfy طلبات HTTP نيابة عن الـ origin server. origin server هو السيرفر الأصلي، mail و DNS ملهمش علاقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5

Web Caching | Proxy server | **ترجيح: high**

.13

A Web cache satisfies HTTP requests on the behalf of what?

ترجمة: الـ Web cache بيـ satisfy طلبات HTTP نيابة عن إيه؟

A. A mail server

B. A DNS server

C. A router

D. An origin Web server

الإجابة الصحيحة: D) An origin Web server

ليه صح: نيابة عن الـ origin Web server. لو عنده نسخة من الـ object بيبعته للـ client من غير ما يروح للـ origin.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5

Web Caching | Origin server | **ترجيح: high**

.14

Where does the Web cache keep copies of recently requested objects?

ترجمة: فين الـ Web cache بيخزّن نسخ الـ objects اللي اتطلبت قريب؟

A. In the DNS

B. In the origin server

C. In its own disk storage

D. In the user's browser only

الإجابة الصحيحة: C) In its own disk storage

ليه صح: في disk storage بتاعه. مش في browser ولا في origin server ولا DNS — الـ cache عنده storage خاص بيه.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5

Web Caching | Storage | **ترجيح: high**

.15

What is the first step when a browser requests an object through a Web cache?

ترجمة: إيه أول خطوة لما الـ browser يطلب object عبر Web cache؟

A. The browser sends a UDP packet to the origin server

B. The browser queries DNS for the cache's IP

C. The browser establishes a TCP connection to the Web cache and sends an HTTP request for the object to the Web cache

D. The browser establishes a TCP connection directly to the origin server

الإجابة الصحيحة: C) The browser establishes a TCP connection to the Web cache and sends an HTTP request for the object to the Web cache

ليه صح: الـ browser بيفتح TCP للـ cache وبعت HTTP request له. مش لـ origin مباشرة ولا UDP ولا DNS query.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (4 steps)

Web Caching | Steps | **ترجيح: high**

.16

If the Web cache does NOT have a copy of the requested object, what does it do?

ترجمة: لو الـ Web cache معندوش نسخة من الـ object، بيعمل إيه؟

A. Returns an error to the browser

B. Opens a TCP connection to the origin server, sends an HTTP request, and stores a copy when it receives the object

C. Sends the request to another browser

D. Forwards the request via UDP multicast

الإجابة الصحيحة: B) Opens a TCP connection to the origin server, sends an HTTP request, and stores a copy when it receives the object

ليه صح: يفتح TCP للـ origin server، يبعت HTTP request، يستقبل الـ object، يخزّن نسخة محلياً، ويبعت نسخة للـ browser. مش error ولا broadcast.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (4 steps)

Web Caching | Steps | **ترجيح: high**

.17

A Web cache acts as:

ترجمة: الـ Web cache بيشتغل كـ إيه؟

A. Both a server and a client at the same time

B. Only a client

C. Only a router

D. Only a server

الإجابة الصحيحة: A) Both a server and a client at the same time

ليه صح: server (لما يبعت response للـ browser) و client (لما يبعت request للـ origin server) في نفس الوقت. نقطة مهمة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5

Web Caching | Server/Client | **ترجيح: high**

.18

Web caching has been deployed for which two main reasons?

ترجمة: Web caching اتعمل deployment عشان إيه السببين الأساسيين؟

A. To replace TCP with UDP for faster transport

B. To reduce response time for client requests and to reduce traffic on an institution's access link

C. To eliminate the need for DNS

D. To encrypt HTTP messages and provide security

الإجابة الصحيحة: B) To reduce response time for client requests and to reduce traffic on an institution's access link

ليه صح: تقليل response time و تقليل traffic على access link. التشفير والـ UDP و إلغاء DNS مالهمش علاقة بـ caching.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (two reasons)

Web Caching | Benefits | **ترجيح: high**

.19

In the institutional network example, the access link between the institutional router and the Internet router has a rate of:

ترجمة: في المثال، الـ access link بين راوتر المؤسسة وراوتر الإنترنت سرعته كام؟

A. 1 Mbps

B. 100 Mbps

C. 15 Mbps

D. 1 Gbps

الإجابة الصحيحة: C) 15 Mbps

ليه صح: 15 Mbps. الـ LAN أسرع (100 Mbps). 15 Mbps هي الـ bottleneck اللي بيدخل الـ traffic intensity = 1.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Example | **ترجيح: high**

.20

In the institutional example, the Internet delay is approximately:

ترجمة: في المثال، الـ Internet delay تقريباً كام؟

A. 2 seconds

B. 2 ms

C. 200 ms

D. 20 seconds

الإجابة الصحيحة: A) 2 seconds

ليه صح: ثانيتين (2 seconds). بتستخدم في حساب الـ average response time مع الـ hit rate.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Example | **ترجيح: high**

.21

What is the traffic intensity on the access link in the original (no-cache) example?

ترجمة: إيه الـ traffic intensity على الـ access link في المثال من غير cache؟

A. 1.0

B. 0.4

C. 0.15

D. 0.6

الإجابة الصحيحة: A) 1.0

ليه صح: 15 Mbps / 1 Mbits * req/s. الـ LAN بتطلع 0.15 (مهملة)، بس 1.0 على access link معناها delay كبير بيكبر من غير bound.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Traffic intensity | **ترجيح: high**

.22

When the traffic intensity on a link approaches 1, the delay on that link:

ترجمة: لما الـ traffic intensity تقرب من 1 على link، الـ delay بيعمل إيه؟

A. Decreases linearly

B. Becomes very large and grows without bound

C. Stays constant at a few milliseconds

D. Becomes zero

الإجابة الصحيحة: B) Becomes very large and grows without bound

ليه صح: بيكبر بشكل كبير وبدون bound. ده عكس intensity 0.15 اللي بتعني delay صغير (tens of ms).

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Traffic intensity | **ترجيح: high**

.23

The first solution in the example was to upgrade the access link to 100 Mbps. What does this give?

ترجمة: الحل الأول (ترقية الـ link لـ 100 Mbps) بيدّي إيه؟

A. No change in response time

B. Traffic intensity 0.4, total response time 1.2 seconds

C. Traffic intensity 0.15, total response time roughly 2 seconds (the Internet delay)

D. Traffic intensity 1.0, total response time on the order of minutes

الإجابة الصحيحة: C) Traffic intensity 0.15, total response time roughly 2 seconds (the Internet delay)

ليه صح: intensity 0.15 (مهملة) + Internet delay 2 seconds = إجمالي ~2 seconds. بس مكلف جداً (upgrade link).

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Upgrade solution | **ترجيح: high**

.24

The second solution installs a Web cache with a hit rate of 0.4. What is the resulting average delay?

ترجمة: الحل التاني (Web cache hit rate 0.4) بيدّي إيه average delay؟

A. Roughly 2 seconds

B. Roughly 0.01 seconds

C. On the order of minutes

D. Slightly greater than 1.2 seconds

الإجابة الصحيحة: D) Slightly greater than 1.2 seconds

ليه صح: 0.4*0.01 + 0.6*2.01 = 0.004 + 1.206 ≈ 1.21 seconds. أقل من الحل الأول ومش محتاج upgrade.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Cache solution | ترجيح: high

.25

With a cache hit rate of 0.4, the traffic intensity on the access link is reduced from 1.0 to:

ترجمة: مع hit rate 0.4، الـ traffic intensity على الـ access link بتقل من 1.0 لـ كام؟

A. 0.01

B. 0.4

C. 0.15

D. 0.6

الإجابة الصحيحة: D) 0.6

ليه صح: 0.6 — لأن 60% من الـ requests (اللي miss) بتمر على الـ access link. 40% بـ hit على الـ cache وما بتمرش.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching | Traffic intensity | ترجيح: high

.26

CDN companies install what to localize much of the Internet traffic?

ترجمة: شركات CDN بـ install إيه عشان تـ localize الـ traffic؟

A. Many routers only

B. A single central cache

C. Many DNS servers

D. Many geographically distributed caches

الإجابة الصحيحة: D) Many geographically distributed caches

ليه صح: caches موزعة جغرافياً. ده بيخلي الـ traffic قريب من اليوزر. مش cache مركزي ولا DNS.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (CDN paragraph)

CDN Caching **ترجيح: high**

.27

What problem does the conditional GET solve?

ترجمة: الـ conditional GET بيحل إيه مشكلة؟

A. DNS resolution may fail

B. The cached copy of an object may be stale

C. Cookies may be lost

D. TCP connections may be dropped

الإجابة الصحيحة: B) The cached copy of an object may be stale

ليه صح: النسخة المخزنة في الـ cache ممكن تبقى stale (الـ object اتغير على الـ origin server). cookies و TCP و DNS مالهمش علاقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET)

Conditional GET Staleness **ترجيح: high**

.28

An HTTP request message is a conditional GET if it uses the GET method AND:

ترجمة: رسالة HTTP request بتبقى conditional GET لو بتستخدم GET وبتشمل إيه؟

A. Includes a Cookie: header line

B. Includes a User-agent: header line

C. Includes a Set-cookie: header line

D. Includes an If-Modified-Since: header line

الإجابة الصحيحة: D) Includes an If-Modified-Since: header line

ليه صح: شرطين: (1) GET method + (2) If-Modified-Since: header. Set-cookie و Cookie و User-agent شروط تانية مختلفة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET definition)

Conditional GET | If-Modified-Since | **ترجيح: high**

.29

The If-Modified-Since: header line value is taken from which previous server response header?

ترجمة: قيمة If-Modified-Since بتيجي من إيه header في الـ response القديم؟

A. Last-Modified:

B. Content-Length:

C. Set-cookie:

D. Date:

الإجابة الصحيحة: A) Last-Modified:

ليه صح: من Last-Modified: header اللي السيرفر بعت في الـ response الأصلي. ده اللي بيخلي الـ cache يحدد لو الـ object اتغير.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET example)

Conditional GET | Last-Modified | **ترجيح: high**

.30

If the object has NOT been modified since the specified date, the Web server responds with which status code?

ترجمة: لو الـ object ما اتغيرش، السيرفر بيرد بـ أيه status code؟

A. 200 OK

B. 301 Moved Permanently

C. 404 Not Found

D. 304 Not Modified

الإجابة الصحيحة: D) 304 Not Modified

ليه صح: 304 Not Modified. 200 OK معناها السيرفر بيبعت الـ object، 404 معناها مش موجود، 301 معناها اتنقل.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET example)

Conditional GET | 304 Not Modified | ترجيح: high

.31

When a 304 Not Modified response is sent, what is the entity body?

ترجمة: لما السيرفر يبعت 304 Not Modified، الـ entity body فيها إيه؟

A. The full requested object

B. The cookies for the user

C. Empty — the requested object is NOT included

D. A small error message

الإجابة الصحيحة: C) Empty — the requested object is NOT included

ليه صح: فاضية — مفيش object. لو السيرفر بعت الـ object هيضيع bandwidth ويزود الـ response time بدون داعي.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET example)

Conditional GET | Empty body | ترجيح: high

.32

Why is the conditional GET important?

ترجمة: ليه الـ conditional GET مهم؟

A. It removes the need for TCP

B. It speeds up DNS resolution

C. It confirms that a cache has up-to-date objects without re-sending the full object when nothing has changed

D. It encrypts the HTTP request

الإجابة الصحيحة: C) It confirms that a cache has up-to-date objects without re-sending the full object when nothing has changed

ليه صح: بيـ verify الـ cache up-to-date من غير إعادة إرسال الـ object لو ما اتغيرش — بيوفّر bandwidth و delay.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET)

Conditional GET | Benefits | **ترجيح: high**

.33

When was HTTP/2 standardized?

ترجمة: متى HTTP/2 اتعمل standardization؟

A. 2020

B. 2015

C. 2005

D. 1997

الإجابة الصحيحة: B) 2015

ليه صح: HTTP/1.1 اتـ standard في 1997. 2015 (RFC 7540) نقطة مهمة في بداية القسم 2.2.6.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6

HTTP/2 | History | **ترجيح: high**

.34

What was the first new version of HTTP since HTTP/1.1?

ترجمة: إيه أول نسخة جديدة من HTTP بعد HTTP/1.1؟

A. HTTPS

B. HTTP/3

C. HTTP/2

D. QUIC

الإجابة الصحيحة: C) HTTP/2

ليه صح: HTTP/3 (2015). HTTP/2 لسه بعده. HTTPS مش نسخة، ده HTTP مع TLS. QUIC بروتوكول نقل مش HTTP.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6

HTTP/2 Versions **ترجيح: high**

.35

As of 2020, what percentage of the top 10 million websites supported HTTP/2?

ترجمة: حتى 2020، كام نسبة من أكبر 10 مليون موقع بتـ support HTTP/2؟

A. Over 40%

B. Exactly 100%

C. Over 90%

D. Under 10%

الإجابة الصحيحة: A) Over 40%

ليه صح: أكتر من 40%. نقطة مهمة في المحاضرة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6

HTTP/2 Adoption **ترجيح: high**

.36

Which is NOT a primary goal of HTTP/2?

ترجمة: إيه اللي مش هدف أساسي من HTTP/2؟

A. Provide request prioritization and server push

B. Reduce perceived latency via request/response multiplexing over a single TCP connection

C. Provide efficient compression of HTTP header fields

D. Replace HTTP/1.1 status codes and URLs

الإجابة الصحيحة: D) Replace HTTP/1.1 status codes and URLs

ليه صح: HTTP/2 ما بياخدش status codes ولا URLs — دي بتفل كما هي. الأهداف: multiplexing + prioritization + server push + header compression.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (primary goals paragraph)

HTTP/2 Goals **ترجيح: high**

.37

What does HTTP/2 NOT change?

ترجمة: HTTP/2 ما بياخدش إيه؟

A. HTTP methods, status codes, URLs, or header fields

B. The default port number

C. The transport protocol used

D. How data is formatted and transported between client and server

الإجابة الصحيحة: A) HTTP methods, status codes, URLs, or header fields

ليه صح: methods و status codes و URLs و headers بتفل كما هي. بس HTTP/2 بياخد إزاي الـ data متformat ومتنقلة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6

HTTP/2 Compatibility **ترجيح: high**

.38

HTTP/2 changes which aspect of HTTP/1.1?

ترجمة: HTTP/2 بياخد إيه في HTTP/1.1؟

A. URL structure

B. Status codes (200, 404, etc.)

C. HTTP methods (GET, POST, etc.)

D. How the data is formatted and transported between the client and server

الإجابة الصحيحة: D) How the data is formatted and transported between the client and server

ليه صح: إزاي الـ data متـ format ومتنقل (framing + interleaving + binary). methods و codes و URLs بتفل كما هي.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6

HTTP/2 Changes **ترجيح: high**

.39

What problem arises when sending all objects of a Web page over a single HTTP/1.1 TCP connection?

ترجمة: إيه المشكلة لما نبعت كل objects صفحة ويب على TCP connection واحد في HTTP/1.1؟

A. DNS resolution failure

B. Head of Line (HOL) blocking

C. Cookie overflow

D. TLS handshake error

الإجابة الصحيحة: B) Head of Line (HOL) blocking

ليه صح: HOL blocking — فيديو كبير على رأس الصف بيـ block الـ small objects وراه. cookies و DNS و TLS مالهمش علاقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HOL blocking paragraph)

HTTP/2 HOL blocking **ترجيح: high**

.40

In HOL blocking, which object blocks the small objects behind it?

ترجمة: في HOL blocking، إيه اللي بـ block الـ small objects وراه؟

A. A DNS query

B. A small image file

C. A cookie file

D. A large video clip at the head of the line on a low-to-medium speed bottleneck link

الإجابة الصحيحة: D) A large video clip at the head of the line on a low-to-medium speed bottleneck link

ليه صح: فيديو كبير على رأس الصف على bottleneck link. small objects بتستنى وراه.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HOL blocking example)

HTTP/2 HOL blocking **ترجيح: high**

.41

How do HTTP/1.1 browsers typically work around the HOL blocking problem?

ترجمة: إزاي HTTP/1.1 browsers بيتعاملوا مع HOL blocking؟

A. By disabling cookies

B. By opening multiple parallel TCP connections

C. By sending all requests over UDP

D. By switching to HTTP/3

الإجابة الصحيحة: B) By opening multiple parallel TCP connections

ليه صح: بفتح TCP connections متوازية. ده بيخلي small objects تيجي بسرعة.

UDP و HTTP/3 و cookies مالهمش علاقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/1.1 workaround)

HTTP/2 HTTP/1.1 Parallel connections **ترجيح: high**

.42

Why does TCP congestion control give browsers an unintended incentive to use multiple parallel TCP connections?

ترجمة: ليه TCP congestion control بيدي browsers حافز لفتح parallel TCP connections؟

A. TCP congestion control disables itself for parallel connections

B. UDP is faster than TCP for small objects

C. TCP requires authentication before opening parallel connections

D. Each TCP connection sharing a bottleneck link gets approximately 1/n of the bandwidth, so opening n connections lets the browser grab a larger portion

الإجابة الصحيحة: D) Each TCP connection sharing a bottleneck link gets approximately 1/n of the bandwidth, so opening n connections lets the browser grab a larger portion

ليه صح: n TCP connections على bottleneck بياخدوا 1/n لكل واحد — فلو فتحت connection كذا تقدر تـ grab أكتر bandwidth (cheat). مش عشان UDP ولا authentication.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (TCP congestion control incentive)

HTTP/2 TCP congestion control **ترجيح: high**

.43

What is the HTTP/2 solution to HOL blocking?

ترجمة: إيه حل HTTP/2 لمشكلة HOL blocking؟

A. Open more TCP connections per Web page

B. Send all objects sequentially without interleaving

C. Use UDP instead of TCP

D. Break each message into small frames and interleave the request and response messages on the same TCP connection

الإجابة الصحيحة: D) Break each message into small frames and interleave the request and response messages on the same TCP connection

ليه صح: break message لـ frames صغيرة و interleave على نفس TCP connection. مش parallel connections ولا UDP ولا sequential.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/2 Framing)

HTTP/2 Framing Interleaving **ترجيح: high**

.44

In the framing example (1 large video of 1000 frames + 8 small objects of 2 frames each), how many frames are sent before all small objects are sent with interleaving?

ترجمة: في مثال framing (فيديو 1000 frame + 8 objects كل واحد 2 frames)، مع interleaving بتبعت كام frame قبل ما كل small objects تتبعت؟

A. 1000 frames

B. 18 frames

C. 8 frames

D. 1016 frames

الإجابة الصحيحة: B) 18 frames

ليه صح: frame 18: بعد frame 1 من الفيديو، أول frames لكل object (8)؛ بعد frame 2 من الفيديو، آخر frames لكل object (8) → 1+8+1+8=18. من غير interleaving 1016 frame.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (framing example)

HTTP/2 Framing Interleaving **ترجيح: high**

.45

In the framing example, without interleaving, how many frames would be sent before all small objects are sent?

ترجمة: في نفس المثال، من غير interleaving، كام frame هتتبعت قبل كل small objects؟

A. 1016 frames

B. 18 frames

C. 1000 frames

D. 2000 frames

الإجابة الصحيحة: A) 1016 frames

ليه صح: frame 1016: الفيديو 1000 frame + كل small object 2 frame = 1000 + 16 = 1016. مع interleaving: 18 frame بس.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (framing example)

HTTP/2 Framing **ترجيح: high**

.46

What is described as the single most important enhancement of HTTP/2?

ترجمة: إيه أهم enhancement في HTTP/2؟

A. The Last-Modified: header

B. The ability to break down an HTTP message into independent frames, interleave them, and reassemble them on the other end

C. Conditional GET

D. Cookies

الإجابة الصحيحة: B) The ability to break down an HTTP message into independent frames, interleave them, and reassemble them on the other end

ليه صح: framing + interleaving + reassembly. ده اللي بيخلي multiplexing ممكن ويقلل user-perceived delay بشكل كبير.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/2 Framing)

HTTP/2 Framing **ترجيح: high**

.47

Why does the framing sublayer also binary-encode the frames?

ترجمة: ليه framing sublayer بـ binary encode الـ frames؟

A. Binary protocols are more efficient to parse, lead to slightly smaller frames, and are less error-prone

B. Binary encoding is more secure than text

C. Binary encoding is required by TCP

D. Binary encoding removes the need for TLS

الإجابة الصحيحة: A) Binary protocols are more efficient to parse, lead to slightly smaller frames, and are less error-prone

ليه صح: binary أكفأ في الـ parsing، أصغر، وأقل خطأ. مش شرط من TCP ولا أأمن ولا بيستبدل TLS.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (framing sublayer)

HTTP/2 Binary encoding **ترجيح: high**

.48

In HTTP/2 response prioritization, what weight range can a client assign to each message?

ترجمة: في HTTP/2 prioritization، نطاق الـ weight كام؟

A. Between 1 and 1024

B. Between 1 and 10

C. Between 1 and 256

D. Between 0 and 100

الإجابة الصحيحة: C) Between 1 and 256

ليه صح: 1 لـ 256. رقم أكبر = أولوية أعلى. السيرفر بيبعت frames الأولى للـ responses ذات الأولوية الأعلى.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Response Message Prioritization)

HTTP/2 Prioritization **ترجيح: high**

.49

In HTTP/2 prioritization, a higher weight number indicates:

ترجمة: في HTTP/2 prioritization، رقم weight أكبر معناه إيه؟

A. Slower response time

B. Lower priority

C. Larger object size

D. Higher priority

الإجابة الصحيحة: D) Higher priority

ليه صح: أولوية أعلى. السيرفر بيبعت frames الأولى للـ responses ذات الأولوية الأعلى.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Response Message Prioritization)

HTTP/2 Prioritization **ترجيح: high**

.50

Which HTTP/2 feature allows a server to send multiple responses for a single client request?

ترجمة: إيه ميزة HTTP/2 اللي بتسمح للسيرفر يبعت كذا response لطلب واحد؟

A. Server push
B. Cookies
C. Last-Modified
D. Conditional GET

الإجابة الصحيحة: A) Server push

ليه صح: Server push. السيرفر بيشوف الـ HTML وبيـ push الـ objects اللي اليوزر محتاجها قبل ما يطلبها. conditional GET و cookies مالهمش علاقة.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Server Push)

HTTP/2 Server push **ترجيح: high**

.51

How does the server know which objects to push in HTTP/2?

ترجمة: إزاي السيرفر بيعرف إيه objects اللي هـ push في HTTP/2؟

A. The cookies tell the server what to push
B. The DNS provides the list of objects
C. The HTML base page indicates the objects that will be needed to fully render the Web page
D. The client explicitly requests each pushed object

الإجابة الصحيحة: C) The HTML base page indicates the objects that will be needed to fully render the Web page

ليه صح: الـ HTML base page بيقول إيه objects اللي محتاجة. السيرفر بيشوفها ويـ push قبل ما اليوزر يطلب. مش الـ client ولا DNS ولا cookies.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Server Push)

HTTP/2 Server push **ترجيح: high**

.52

What does server push eliminate?

ترجمة: Server push بيـ eliminate إيه؟

A. The need for cookies

B. The need for TCP connections

C. The extra latency due to waiting for the requests

D. The need for DNS

الإجابة الصحيحة: C) The extra latency due to waiting for the requests

ليه صح: latency اللي بيتسبب عن استنى الطلبات. السيرفر بيبعت objects قبل ما اليوزر يطلبها. مش cookies ولا TCP ولا DNS.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Server Push)

HTTP/2 Server push **ترجيح: high**

.53

HTTP/3 is designed to operate over which transport protocol?

ترجمة: HTTP/3 متصمم عشان يشتغل فوق إيه؟

A. QUIC

B. UDP directly

C. SCTP

D. TCP

الإجابة الصحيحة: A) QUIC

ليه صح: QUIC. مش TCP ولا UDP مباشرة. QUIC بروتوكول نقل جديد.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 QUIC **ترجيح: high**

.54

QUIC is a new "transport" protocol implemented in which layer over the bare-bones UDP protocol?

ترجمة: QUIC بروتوكول نقل جديد متـ implement في أي طبقة فوق UDP؟

A. The application layer

B. The network layer

C. The physical layer

D. The transport layer

الإجابة الصحيحة: A) The application layer

ليه صح: application layer فوق UDP. ده ملاحظة مهمة — QUIC مش في transport layer بتاع OS، ده application-level.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 QUIC Application layer **ترجيح: high**

.55

QUIC is implemented over which underlying protocol?

ترجمة: QUIC متـ implement فوق إيه؟

A. IP

B. UDP

C. TLS

D. TCP

الإجابة الصحيحة: B) UDP

ليه صح: UDP bare-bones. ده اللي بيخليه أسرع وأقدر يـ multiplex من غير head-of-line blocking بتاع TCP.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 QUIC UDP **ترجيح: high**

.56

Which desirable features does QUIC have for HTTP?

ترجمة: QUIC فيه إيه مميزات مرغوبة لـ HTTP؟

A. Encryption, authentication, and routing only

B. File transfer, e-mail, and DNS

C. Cookies, conditional GET, and DNS

D. Message multiplexing (interleaving), per-stream flow control, and low-latency connection establishment

الإجابة الصحيحة: D) Message multiplexing (interleaving), per-stream flow control, and low-latency connection establishment

ليه صح: message multiplexing، per-stream flow control، low-latency connection establishment. النقاط دي بتسمح لـ HTTP/3 بـ design أبسط.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 QUIC Features **ترجيح: high**

.57

Because QUIC subsumes many HTTP/2 features, what does HTTP/3's design become?

ترجمة: بما إن QUIC بـ subsume مميزات HTTP/2، HTTP/3 بيبقى إزاي؟

A. More complex and slower

B. Simpler and more streamlined

C. Identical to HTTP/1.1

D. Backward-compatible with HTTP/0.9

الإجابة الصحيحة: B) Simpler and more streamlined

ليه صح: أبسط وأكتر streamlined. مش أكتر تعقيد، ولا متوافق مع HTTP/0.9، ولا زي HTTP/1.1.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 QUIC Design **ترجيح: high**

.58

Cookies are defined in which RFC?

ترجمة: Cookies متعرفة في RFC كام؟

A. RFC 6265

B. RFC 7230

C. RFC 7540

D. RFC 7232

الإجابة الصحيحة: A) RFC 6265

ليه صح: RFC 6265. RFC 7230 لـ HTTP/1.1، RFC 7540 لـ HTTP/2، RFC 7232 لـ conditional GET.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies RFC **ترجيح: medium**

.59

Typically, who purchases and installs a Web cache?

ترجمة: مين بيـ install الـ Web cache عادة؟

A. The origin Web server

B. An ISP (for example, a university on its campus network)

C. The DNS registry

D. The end user on a personal computer

الإجابة الصحيحة: B) An ISP (for example, a university on its campus network)

ليه صح: ISP (مثل جامعة على campus network). اليوزر مش بيـ install الـ cache بنفسه، ولا الـ origin server.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5

Web Caching ISP **ترجيح: medium**

.60

In the institutional example, what is the average object size?

ترجمة: في المثال، إيه حجم الـ object المتوسط؟

A. 10 Mbits

B. 1 Mbits

C. 100 Kbits

D. 1 Kbits

الإجابة الصحيحة: B) 1 Mbits

ليه صح: 1 Mbits. الـ request rate 15 req/s، فالـ traffic intensity = 15*1/15 = 1 على الـ access link.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12 example)

Web Caching Example **ترجيح: medium**

.61

In practice, hit rates for Web caches typically range from:

ترجمة: في الواقع، الـ hit rates للـ Web caches بتتراوح بين كام؟

A. 0.5 to 0.9

B. 0.9 to 1.0

C. 0.2 to 0.7

D. 0.01 to 0.1

الإجابة الصحيحة: C) 0.2 to 0.7

ليه صح: 0.2 لـ 0.7. القيم دي معقولة للـ caches في المؤسسات.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (hit rate paragraph)

Web Caching Hit rate **ترجيح: medium**

.62

The conditional GET is defined in which RFC?

ترجمة: الـ conditional GET متعرف في RFC كام؟

A. RFC 7232

B. RFC 1945

C. RFC 6265

D. RFC 7540

الإجابة الصحيحة: A) RFC 7232

ليه صح: RFC 7232 لـ cookies، RFC 7540 لـ HTTP/2، RFC 1945 لـ HTTP/1.0. RFC 6265.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET)

Conditional GET | RFC | **ترجيح: medium**

.63

How many parallel TCP connections do many HTTP/1.1 browsers typically open?

ترجمة: كم TCP connection متوازي بيفتح HTTP/1.1 browser؟

A. Up to two

B. Up to ten

C. Up to one hundred

D. Up to six

الإجابة الصحيحة: D) Up to six

ليه صح: لـ 6 parallel TCP connections. عشان يتجاوزوا HOL ويـ grab أكتر bandwidth.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6

HTTP/2 | HTTP/1.1 | Parallel connections | **ترجيح: medium**

.64

As of 2020, HTTP/3 was described in what status?

ترجمة: حتى 2020، HTTP/3 كان في أيه status؟

A. Internet drafts, not yet fully standardized

B. Proprietary to Google only

C. Already obsolete

D. Fully standardized RFC

الإجابة الصحيحة: A) Internet drafts, not yet fully standardized

ليه صح: Internet drafts — لسه مش متـ standardize بالكامل. مش RFC نهائي، ولا obsolete، ولا proprietary.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 Status **ترجيح: medium**

Part 2 — Essay Questions

Essay 1.

Define cookies and explain why HTTP uses them despite the server being stateless.

ترجمة: عرّف cookies واشرح ليه HTTP بيستخدمها رغم إن السيرفر stateless.

HTTP is a stateless protocol: the HTTP server maintains no information about clients, which simplifies server design and allows high-performance Web servers to handle thousands of simultaneous TCP connections.
However, it is often desirable for a Web site to identify users, either to restrict user access or to serve content as a function of the user identity.
For these purposes, HTTP uses cookies (defined in RFC 6265).
Cookies allow sites to keep track of users — most major commercial Web sites use cookies today.
So cookies add per-user state on top of the otherwise-stateless HTTP protocol, while keeping the core protocol simple.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4

Cookies Stateless Definition **ترجيح: high**

Essay 2.

List the four components of cookie technology and briefly explain the role of each.

ترجمة: اكتب المكونات الأربعة لتقنية cookies واشرح دور كل واحد باختصار.

Cookie technology has four components (Figure 2.10):
1. A cookie header line in the HTTP response message — the server sends this to give the browser a unique identification number (e.g., Set-cookie: 1678).
2. A cookie header line in the HTTP request message — the browser sends this back to the server on subsequent requests so the server can identify the user (e.g., Cookie: 1678).
3. A cookie file kept on the user's end system and managed by the user's browser — stores the hostname and the identification number for each site the user has visited.
4. A back-end database at the Web site — indexed by the identification number, it stores per-user information (visited pages, shopping cart, registration data, etc.).

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (Figure 2.10)

Cookies Components **ترجيح: high**

Essay 3.

Walk through the Susan/Amazon example to illustrate how cookies work.

ترجمة: اتمم مثال Susan/Amazon عشان توضح إزاي cookies بتشتغل.

1. Susan contacts Amazon.com for the first time from her home PC using Internet Explorer; suppose she has previously visited eBay.
2. When the request reaches the Amazon Web server, the server creates a unique identification number (e.g., 1678) and creates an entry in its back-end database indexed by that number.
3. The Amazon server responds with an HTTP response containing a Set-cookie: 1678 header.
4. Susan's browser receives the response, sees the Set-cookie: header, and appends a line to its cookie file including the Amazon hostname and the number 1678 (the file already has an eBay entry from her past visit).
5. As Susan continues browsing Amazon, each subsequent request includes the header line Cookie: 1678, allowing Amazon to track which pages user 1678 visited, in which order, and at what times.
6. Amazon uses this for its shopping cart service (collecting purchases for later combined checkout) and for product recommendations.
7. If Susan registers (name, e-mail, address, credit card), Amazon associates that information with the number 1678 in its database — enabling 'one-click shopping'.
8. When Susan returns a week later, her browser still sends Cookie: 1678, so Amazon can identify her again.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (Susan/Amazon example)

Cookies Example **ترجيح: high**

Essay 4.

Discuss the privacy concerns associated with cookies.

ترجمة: ناقش المخاوف الـ privacy المرتبطة بـ cookies.

Although cookies often simplify the Internet shopping experience for the user, they are controversial because they can also be considered as an invasion of privacy.
Using a combination of cookies and user-supplied account information, a Web site can learn a lot about a user — including which pages were visited, in what order, at what times, plus full name, e-mail address, postal address, and credit card information if the user registered.
The site can then potentially sell this information to a third party.
Even without explicit registration, a site that combines cookies across visits can build a detailed profile of user behavior and interests.
This is the core of the privacy debate: cookies are technically just an identification mechanism, but the data they enable to be collected is what raises the privacy concern.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.4 (privacy paragraph)

Cookies Privacy **ترجيح: high**

Essay 5.

Define a Web cache (proxy server) and describe the four steps of how it handles a browser request.

ترجمة: عرّف الـ Web cache (proxy server) ووصّف الخطوات الأربعة لإزاي بيتعامل مع طلب browser.

A Web cache — also called a proxy server — is a network entity that satisfies HTTP requests on the behalf of an origin Web server. The Web cache has its own disk storage and keeps copies of recently requested objects.
Steps when a browser requests an object (e.g., www.someschool.edu/campus.gif):
1. The browser establishes a TCP connection to the Web cache and sends an HTTP request for the object to the Web cache.
2. The Web cache checks if it has a copy of the object stored locally. If it does, the Web cache returns the object within an HTTP response message to the client browser.
3. If the Web cache does NOT have the object, the Web cache opens a TCP connection to the origin server, sends an HTTP request for the object, and the origin server sends the object back to the cache in an HTTP response.
4. When the Web cache receives the object, it stores a copy in its local storage and sends a copy, within an HTTP response message, to the client browser (over the existing TCP connection between the client and the cache).

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (4 steps)

Web Caching Steps **ترجيح: high**

Essay 6.

Explain why a Web cache can act as both a server and a client at the same time.

ترجمة: اشرح ليه الـ Web cache يقدر يشتغل كـ server و client في نفس الوقت.

A Web cache plays two roles depending on which side it is talking to:
- When it receives requests from a browser and sends responses back to the browser, it acts as a server.
- When it sends requests to the origin server and receives responses from the origin server, it acts as a client.

These two roles are interleaved within a single user request: first the cache is a server toward the browser, then (on a cache miss) it becomes a client toward the origin server, and finally it is again a server toward the browser when delivering the object.

This dual role is exactly what allows the cache to satisfy the browser's request without the browser having to talk to the origin server directly.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5

Web Caching | Server/Client | **ترجيح: high**

Essay 7.

State the two main reasons Web caching has been deployed in the Internet.

ترجمة: اكتب السببين الأساسيين لـ deployment الـ Web caching في الإنترنت.

Web caching has been deployed for two main reasons:
1. Reduced response time for client requests — particularly when the bottleneck bandwidth between the client and the origin server is much less than the bottleneck bandwidth between the client and the cache. If there is a high-speed connection between the client and the cache and the cache has the requested object, the cache can deliver the object rapidly.
2. Reduced traffic on an institution's access link to the Internet — by reducing traffic, the institution does not have to upgrade bandwidth as quickly, reducing costs. Furthermore, Web caches can substantially reduce Web traffic in the Internet as a whole, improving performance for all applications.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (two reasons)

Web Caching | Benefits | **ترجيح: high**

Essay 8.

Define a conditional GET and list the two conditions that make an HTTP request a conditional GET.

ترجمة: عرّف الـ conditional GET واكتب الشرطين اللي بيخلوها conditional GET.

A conditional GET is an HTTP mechanism that allows a cache to verify that its cached objects are up to date without re-downloading the full object if it has not changed.
An HTTP request message is a conditional GET if BOTH:
1. The request message uses the GET method.
2. The request message includes an If-Modified-Since: header line.
If the object has not been modified since the specified date, the server returns a 304 Not Modified response with an empty entity body, telling the cache that its local copy is still valid. This saves bandwidth and reduces user-perceived response time, especially for large objects.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET)

Conditional GET | Definition | **ترجيح: high**

Essay 9.

Walk through the conditional GET example with the kiwi.gif object.

ترجمة: اتمم مثال الـ conditional GET مع الـ object الـ kiwi.gif.

First, on behalf of a requesting browser, the proxy cache sends a request to the Web server:

```
GET /fruit/kiwi.gif HTTP/1.1
Host: www.exotiquecuisine.com
```

Second, the Web server responds with the object plus a Last-Modified: header:

```
HTTP/1.1 200 OK
Last-Modified: Wed, 9 Sep 2015 09:23:24
Content-Type: image/gif
(data ...)
```

The cache forwards the object to the browser but also stores it locally along with the Last-Modified date.

Third, one week later, another browser requests the same object via the cache and the object is still in the cache. The cache issues a conditional GET:

```
GET /fruit/kiwi.gif HTTP/1.1
Host: www.exotiquecuisine.com
If-modified-since: Wed, 9 Sep 2015 09:23:24
```

The value of If-Modified-Since is exactly the Last-Modified value the server sent one week ago.

Fourth, if the object has NOT been modified since 9 Sep 2015 09:23:24, the server responds:

```
HTTP/1.1 304 Not Modified
(empty entity body)
```

The 304 status tells the cache it can go ahead and forward its cached copy to the requesting browser. The object itself is not re-sent, saving bandwidth.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Conditional GET example)

Conditional GET | Example | **ترجيح: high**

Essay 10.

Explain the Head of Line (HOL) blocking problem in HTTP/1.1 and how HTTP/1.1 browsers work around it.

ترجمة: اشرح مشكلة الـ Head of Line blocking في HTTP/1.1 وإزاي browsers بيتعاملوا معاها.

HOL blocking arises because HTTP/1.1 uses persistent TCP connections, sending all objects of a Web page over a single TCP connection.
Consider a Web page with an HTML base page, a large video clip near the top, and many small objects below the video. With a low-to-medium speed bottleneck link on the path between server and client, the video clip takes a long time to pass through the bottleneck link, while the small objects behind it must wait — the video clip at the head of the line blocks the small objects behind it.

HTTP/1.1 browsers typically work around this by opening multiple parallel TCP connections, sending objects in parallel. This way the small objects can arrive and render faster, reducing user-perceived delay.
TCP congestion control gives browsers an unintended incentive for this: roughly, n TCP connections sharing a bottleneck each get 1/n of the bandwidth, so opening more connections lets a browser grab a larger portion. Many HTTP/1.1 browsers open up to six parallel TCP connections — both to circumvent HOL blocking and to obtain more bandwidth.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HOL blocking)

HTTP/2 HTTP/1.1 HOL blocking **ترجيح: high**

Essay 11.

Explain the primary goals of HTTP/2 and what HTTP/2 does NOT change.

ترجمة: اشرح أهداف HTTP/2 الأساسية وإيه اللي ما بياخدش.

The primary goals of HTTP/2 are:
1. Reduce perceived latency by enabling request and response multiplexing over a single TCP connection.
2. Provide request prioritization.
3. Provide server push.
4. Provide efficient compression of HTTP header fields.

What HTTP/2 does NOT change:
- HTTP methods (GET, POST, HEAD, PUT, DELETE, etc.) — same as before.
- Status codes (200, 301, 404, 505, etc.) — same as before.
- URLs — same as before.
- Header fields — same as before.

Instead, HTTP/2 changes how the data is formatted and transported between the client and server (framing + interleaving + binary encoding). This keeps existing HTTP applications compatible while improving performance.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (primary goals paragraph)

HTTP/2 Goals Compatibility **ترجيح: high**

Essay 12.

Explain HTTP/2 framing and how interleaving solves HOL blocking.

ترجمة: اشرح HTTP/2 framing وإزاي interleaving بيحل مشكلة HOL blocking.

HTTP/2's solution for HOL blocking is to break each HTTP message into small frames and interleave the request and response messages on the same TCP connection.
Example: a Web page with one large video clip (1000 frames) and 8 smaller objects (2 frames each). The server receives 9 concurrent requests and needs to send 9 competing responses.
With frame interleaving: after sending one frame of the video clip, the first frame of each of the 8 small objects is sent; then the second frame of the video clip, then the second (last) frame of each small object. So all 8 small objects are fully sent after only 18 frames total.
Without interleaving, the smaller objects would only be sent after sending 1016 frames (1000 video + 16 small-objects), so the user would wait much longer to see any of the small objects render.
The framing sublayer also binary-encodes the frames — binary protocols are more efficient to parse, lead to slightly smaller frames, and are less error-prone.
This ability to break down an HTTP message into independent frames, interleave them, and reassemble them on the other end is the single most important enhancement of HTTP/2.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/2 Framing)

HTTP/2 Framing Interleaving **ترجيح: high**

.Essay 13

Explain HTTP/2 response prioritization.

ترجمة: اشرح HTTP/2 response prioritization.

Message prioritization allows developers to customize the relative priority of requests to better optimize application performance.
The framing sub-layer organizes messages into parallel streams of data destined to the same requestor.
When a client sends concurrent requests to a server, it can prioritize the responses it is requesting by assigning a weight between 1 and 256 to each message. A higher number indicates higher priority.
Using these weights, the server can send first the frames for the responses with the highest priority, so that the most important content reaches the user as early as possible.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Response Message Prioritization)

HTTP/2 Prioritization **ترجيح: high**

Essay 14.

Explain HTTP/2 server push and how it reduces latency.

ترجمة: اشرح HTTP/2 server push وإزاي بيقلل latency.

Server push is an HTTP/2 feature that allows a server to send multiple responses for a single client request. In addition to the response to the original request, the server can push additional objects to the client without the client having to request each one.
This is possible because the HTML base page indicates the objects that will be needed to fully render the Web page. Instead of waiting for HTTP requests for these objects, the server analyzes the HTML page, identifies the objects that are needed, and sends them to the client before receiving explicit requests for them.
Server push eliminates the extra latency that would be caused by waiting for the requests to arrive — the user's browser can start rendering the additional objects as soon as they are pushed, without first having to issue and wait for separate requests.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (Server Push)

HTTP/2 Server push **ترجيح: high**

Essay 15.

Explain HTTP/3 and QUIC, and how HTTP/3 differs from HTTP/2.

ترجمة: اشرح HTTP/3 و QUIC، وإزاي HTTP/3 بيختلف عن HTTP/2.

HTTP/3 is a new HTTP protocol designed to operate over QUIC. As of 2020, HTTP/3 was described in Internet drafts and had not yet been fully standardized.
QUIC is a new 'transport' protocol that is implemented in the application layer over the bare-bones UDP protocol (i.e., not in the OS transport layer like TCP).
QUIC has several features that are desirable for HTTP:
- Message multiplexing (interleaving) — like HTTP/2's interleaving but at the transport level.
- Per-stream flow control.
- Low-latency connection establishment.
Because QUIC subsumes many of the HTTP/2 features (such as message interleaving), HTTP/3 can have a simpler, streamlined design — much of what HTTP/2 had to do at the application layer is handled by QUIC at the transport layer. HTTP/3 over QUIC also avoids TCP's head-of-line blocking problem.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (HTTP/3)

HTTP/3 QUIC **ترجيح: high**

Essay 16.

Compare HTTP/1.1, HTTP/2, and HTTP/3 in terms of transport protocol and key features.

ترجمة: قارن HTTP/1.1 و HTTP/2 و HTTP/3 من حيث بروتوكول النقل والميزات الأساسية.

HTTP/1.1 (RFC 7230, standardized 1997):
- Transport: TCP.
- Persistent connections by default.
- Pipelining allowed but rarely used because of HOL blocking.
- Browsers typically open multiple parallel TCP connections (up to ~6) to circumvent HOL blocking and to grab more bandwidth.

HTTP/2 (RFC 7540, standardized 2015):
- Transport: TCP.
- Same methods, status codes, URLs, and headers as HTTP/1.1.
- Framing + interleaving on a single TCP connection to solve HOL blocking.
- Binary-encoded frames (more efficient parsing, smaller frames, less error-prone).
- Response prioritization (weight 1–256).
- Server push.
- Header compression (HPACK).

HTTP/3 (over QUIC, drafts as of 2020):
- Transport: QUIC over UDP (application-layer transport).
- Solves TCP-level HOL blocking by using QUIC streams.
- Per-stream flow control and low-latency connection establishment.
- Encryption built into QUIC by default.
- Simpler design than HTTP/2 because many HTTP/2 features (e.g., interleaving) are subsumed by QUIC.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (whole)

HTTP/1.1 HTTP/2 HTTP/3 Comparison **ترجيح: high**

Part 3 — Calculation Questions

Calculation 1.

Compare the two solutions to the institutional network bottleneck problem: upgrading the access link vs installing a Web cache.

ترجمة: قارن بين حلّين لمشكلة الـ bottleneck في الشبكة المؤسسية: ترقية الـ access link مقابل install Web cache.

Setup: institutional LAN (high-speed), 15 Mbps access link to the Internet, average object size 1 Mbits, request rate 15 req/s, Internet delay 2 seconds.
Without any change, traffic intensity on access link = 15*1/15 = 1.0, so delay on the link grows without bound and average response time is on the order of minutes.

Solution 1 — Upgrade access link to 100 Mbps:
- Traffic intensity drops to 0.15 (negligible delay).
- Total response time ≈ Internet delay ≈ 2 seconds.
- Cost: institution must pay to upgrade the access link (expensive).

Solution 2 — Install a Web cache (no link upgrade):
- Cache hit rate 0.4: 40% of requests satisfied in ~10 ms by the cache (over the high-speed LAN).
- The remaining 60% go to origin servers, so traffic intensity on access link drops from 1.0 to 0.6 (small delay, tens of milliseconds).
- Average delay = 0.4 * 0.01 + 0.6 * 2.01 ≈ 1.21 seconds.
- Cost: only the cost of purchasing and installing a Web cache (low).

Conclusion: Solution 2 gives an even lower response time (1.21 s vs ~2 s) at a much lower cost — caching wins on both metrics.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12/2.13 example)

Web Caching | Comparison | Calculation reasoning | **ترجيح: high**

Calculation 2.

(Calculation) Compute the average response time for the institutional network example with a Web cache hit rate of 0.4. Given: cache hit delay ≈ 0.01 s (10 ms) for 40% of requests, miss delay ≈ 2.01 s (2 s Internet delay + small access-link delay) for 60% of requests.

ترجمة: (مسألة) احسب إجمالي average response time في مثال المؤسسة مع hit rate 0.4.
المعطيات: cache hit delay ≈ 0.01 s لـ 40% من الطلبات، miss delay ≈ 2.01 s لـ 60% من الطلبات.

Given:
- Cache hit rate = 0.4 (40% of requests).
- Cache hit delay ≈ 0.01 s (objects served locally over the high-speed LAN).
- Cache miss rate = 0.6 (60% of requests still go to the origin server).
- Cache miss delay ≈ 2.01 s (small access-link delay + 2 s Internet delay).

Formula: Average response time = (hit rate * hit delay) + (miss rate * miss delay).

Calculation:
Average response time = 0.4 * 0.01 + 0.6 * 2.01
= 0.004 + 1.206
= 1.21 seconds.

Answer: about 1.21 seconds (slightly greater than 1.2 s).

Comparison: with the no-cache solution the access link intensity is 1.0 → delay grows without bound → on the order of minutes; with the upgrade-to-100 Mbps solution the total is ~2 s. So the cache solution is faster AND cheaper.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12/2.13 example)

Calculation | Web Caching | Hit rate **ترجيح: high**

Calculation 3.

(Calculation) Compute the traffic intensity on the access link in the institutional example if the cache hit rate is 0.4. Given: 15 requests/s, 1 Mbits per object, 15 Mbps access link.

ترجمة: (مسألة) احسب الـ traffic intensity على الـ access link لو hit rate = 0.4. المعطيات: 15 req/s، 1 Mbits/object، 15 Mbps access link.

Given:
- Request rate = 15 requests/s.
- Average object size = 1 Mbits.
- Access link rate = 15 Mbps.
- Cache hit rate = 0.4 → 40% served by cache (do not use access link), 60% go to origin (use access link).

Without cache:
Traffic intensity = (15 req/s * 1 Mbits/req) / 15 Mbps = 15 / 15 = 1.0.

With cache hit rate 0.4, only 60% of requests traverse the access link:
Effective request rate on access link = 0.6 * 15 = 9 req/s.
Traffic intensity = (9 req/s * 1 Mbits/req) / 15 Mbps = 9 / 15 = 0.6.

Answer: 0.6 — a traffic intensity less than 0.8 corresponds to a small delay (tens of milliseconds), which is negligible compared with the 2-second Internet delay.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12/2.13 example)

Calculation | Web Caching | Traffic intensity | **ترجيح: high**

.Calculation 4

(Calculation) In the HTTP/2 framing example, compare the number of frames needed to send all 8 small objects (2 frames each) before sending the large video (1000 frames): (a) with interleaving, (b) without interleaving.

ترجمة: (مسألة) في مثال framing: 8 small objects (كل واحد 2 frames) + فيديو 1000 frames. قارن كام frames لازم عشان كل small objects تتبعت: (a) مع interleaving، (b) من غير interleaving.

Given:
- 1 large video clip = 1000 frames.
- 8 small objects, each = 2 frames (total 16 frames for the small objects).
- All frames are of fixed length.

(a) With interleaving:
Step 1: send 1 frame of the video.
Step 2: send the FIRST frame of each of the 8 small objects (8 frames).
Step 3: send the SECOND frame of the video.
Step 4: send the LAST frame of each of the 8 small objects (8 frames).
Total frames sent before ALL small objects are completely sent = 1 + 8 + 1 + 8 = 18 frames.

(b) Without interleaving (sequential):
All 1000 video frames must be sent first, then the 8 small objects' frames are sent.
Total frames before all small objects are sent = 1000 (video) + 16 (small objects) = 1016 frames.

Answer: 18 frames with interleaving vs 1016 frames without interleaving. HTTP/2 framing decreases user-perceived delay by roughly 1016 / 18 ≈ 56× for the small objects in this example.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.6 (framing example)

Calculation | HTTP/2 | Framing | Interleaving **ترجيح: high**

.Calculation 5

(Calculation) An institution has a 15 Mbps access link, request rate 15 req/s, object size 1 Mbits, Internet delay 2 s. Estimate the average response time if the cache hit rate is 0.6 (instead of 0.4) and the cache hit delay is 0.01 s.

ترجمة: (مسألة) مؤسسة معها 15 Mbps access link، 15 req/s، 1 Mbits/object، Internet delay 2 s. احسب average response time لو hit rate = 0.6 (بدل 0.4) و cache hit delay = 0.01 s.

Given:
- Access link rate = 15 Mbps.
- Request rate = 15 req/s.
- Object size = 1 Mbits.
- Internet delay = 2 s.
- Cache hit rate = 0.6.
- Cache hit delay = 0.01 s.
- Cache miss rate = 0.4.

Step 1 — Traffic intensity on access link with cache:
Only 40% of requests traverse the access link:
Effective request rate on access link = 0.4 * 15 = 6 req/s.
Traffic intensity = (6 * 1) / 15 = 0.4 (well below 0.8 → small delay, tens of ms).

Step 2 — Miss delay ≈ access-link delay (small, say 0.01 s) + Internet delay (2 s) ≈ 2.01 s.

Step 3 — Average response time = hit rate * hit delay + miss rate * miss delay
= 0.6 * 0.01 + 0.4 * 2.01
= 0.006 + 0.804
= 0.81 seconds.

Answer: about 0.81 seconds.

Comparison: with hit rate 0.4 the average was ~1.21 s; raising the hit rate to 0.6 brings the average down to ~0.81 s — a 33% improvement. This shows why improving cache hit rate directly improves user-perceived response time.

المرجع: Lecture 6 (Ch 2, Lec 3) — Section 2.2.5 (Figure 2.12/2.13 example, variant)

Calculation Web Caching Hit rate ترجيح: high